

Lab 4: DFT

Instructions:

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
2. All the tasks in each lab have to be in one main MATLAB file with ".m" extension, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
3. Plots should include x label, y label, legend, and titles
4. File name: ID_Lab4.m/ID_Lab4.rar or zip **and** ID_Lab4.pdf
5. If the report/code is found to be copied, zero marks will be awarded [**Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**]

Note: Do not use the `fft` command to calculate DFT for this lab exercise. Implement DFT using the matrix method discussed in the lecture.

1. Consider the following finite-length sequence.

$$x(n) = \begin{cases} \text{sinc}^2\{(n - 50)/2\}, & 0 \leq n \leq 100; \\ 0, & \text{else.} \end{cases}$$

- a) Determine the DFT $X(k)$ of $x(n)$. Plot (using the stem function) its magnitude and phase. To compute DFT, take N as approximately 3 times the last digit of your ID, If the digit is zero, then add 9, and then take three times the value. [round off to nearest multiple of 2]. Observe for different variations of N , how the DFT output varies, and comment on the results.
 - b) Plot the magnitude and phase of the DTFT $X(e^{j\omega})$ of $x(n)$.
 - c) Verify that the above DFT is the sampled version of $X(e^{j\omega})$. It might be helpful to combine the above two plots in one graph using the **hold** function.
 - d) Make use of the interpolation formula discussed in the lecture to reconstruct the DTFT $X(e^{j\omega})$ from the DFT $X(k)$. Plot your result and comment on the same.
2. Let $X(k)$ denote the N -point DFT of an N -point sequence $x(n)$. The DFT $X(k)$ itself is an N -point sequence. If the DFT of $X(k)$ is computed to obtain another N -point sequence $x_1(n)$, we can show that $x_1(n) = N x((-n))_N$, $0 \leq n \leq N - 1$. Using this property, design a MATLAB function to implement an N -point circular folding operation $x_2(n) = x_1((-n))_N$. The format should be

```
x2 = circfold (x1, N)
% Circular folding using DFT
% x2 = circfold (x1, N)
% x2 = circularly folded output sequence
% x1 = input sequence of length <= N
% N = circular buffer length
```

Determine the circular folding of the sequence: $x_1(n) = \{1, c, 5, 7, 9, -7, -5, -c, -1\}$, c is the second last digit of your ID. If zero, add 3.
